Net Force and Acceleration (template) — teacher key

Question: What decides how quickly something speeds up?

Outline and pacing (42 min)

Tab	Priority	Minutes	If behind
start	1	2	never trimmed
warmup	1	4	never trimmed
predict	1	4	never trimmed
explore	3	8	skipped first
build	2	8	shrinks to 2 of 4
practice	1	8	never trimmed
check	1	4	never trimmed
Huddle h1	clock	minute 18	pulled in by the clock

Targets

- I can say what net force means.
- I can predict how force and mass change acceleration.
- I can calculate acceleration with a = F / m.

Predictions

- A cart is pushed with a steady force. Now the force doubles. What happens to its acceleration? **It doubles.** a = F / m: double F, double a (mass is unchanged).
- The same force pushes a cart. Now the cart carries twice the mass. What happens to its acceleration? **It halves.** a = F / m: double m, half a (force is unchanged).

Huddle script (answers in green)

Definitions

- **Net force:** The **sum** of all the forces on an object.
- **Acceleration:** How fast the **velocity** changes.

Worked example

A 2 kg cart is pushed with a net force of 6 N. Find its acceleration.

- a = F / m
- a = 6 / 2 = **3** m/s^2
- In 1 s its speed rises by **3** m/s.

Label the cart

1. The **applied force** pushes to the right.
2. **Friction** pushes to the left.

Every level, with answers

Find the acceleration

Level 1. A 3 kg cart is pushed with a net force of 9 N. What is its acceleration?
Answer: 3 m/s² a = F / m = 9 / 3 = 3 m/s².

Level 2. A 0.5 kg cart feels a net force of 3 N. What is its acceleration?
Answer: 6 m/s² a = 3 / 0.5 = 6 m/s².

Level 3. A 5 kg cart is pushed right with 15 N while 5 N pushes left. What is its acceleration?
Answer: 2 m/s² Net force = 15 – 5 = 10 N. a = 10 / 5 = 2 m/s².

Level 4. A rope pulls a 6 kg crate with 21 N. Friction is 3 N. What is the acceleration?
Answer: 3 m/s² Net = 21 – 3 = 18 N. a = 18 / 6 = 3 m/s².

Find the force

Level 1. A 3 kg cart accelerates at 3 m/s². What net force acts on it?
Answer: 9 N F = m a = 3 × 3 = 9 N.

Level 2. A 0.5 kg cart accelerates at 4 m/s². What net force acts on it?
Answer: 2 N F = 0.5 × 4 = 2 N.

Level 3. A 2500 g cart accelerates at 2 m/s². What net force acts on it, in newtons?
Answer: 5 N 2500 g = 2.5 kg. F = 2.5 × 2 = 5 N.

Level 4. A 4 kg cart starts from rest and reaches 6 m/s in 2 s. What net force acted on it?
Answer: 12 N a = 6 / 2 = 3 m/s². F = 4 × 3 = 12 N.

Common misconceptions

Students say	Say instead
A bigger force means a bigger speed.	Force sets how fast the speed changes, not the speed itself.
The heavier cart falls or moves faster.	More mass means less acceleration for the same force.
Subtract mass from force.	Divide. Check the units: N / kg = m/s².